

EXPERIMENT 14: DECISION TREE

AIM

To implement a **Decision Tree classification algorithm** in Python and classify data based on given features.

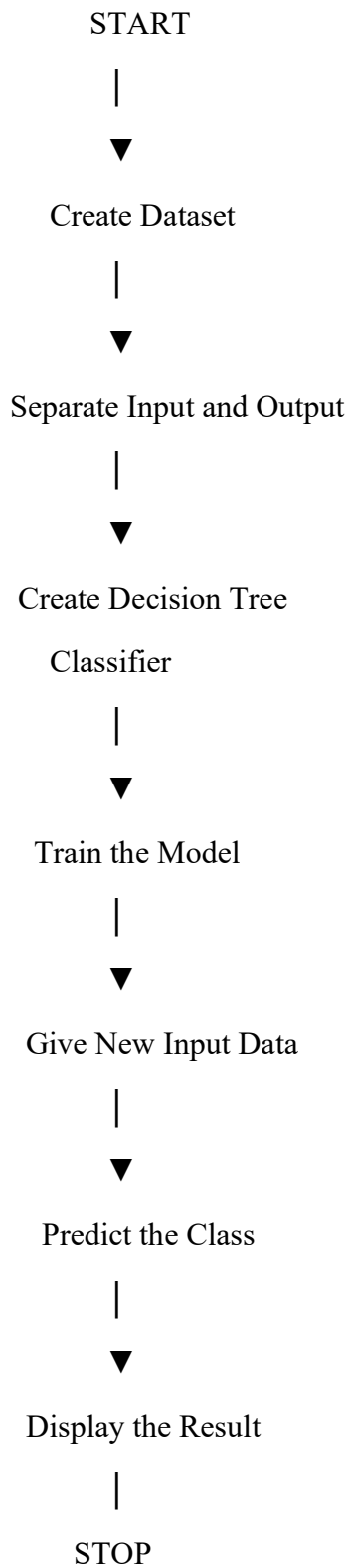
OBJECTIVE

- To understand the concept of Decision Trees.
- To build a classification model using a Decision Tree.
- To train the model using sample data.
- To predict the class of new data.

ALGORITHM

1. Start the program.
2. Import the required libraries.
3. Create the dataset with input features and target values.
4. Separate the input features (X) and target (Y).
5. Create a Decision Tree Classifier.
6. Train the classifier using the dataset.
7. Give a new input for prediction.
8. Predict the class using the trained model.
9. Display the prediction.
10. Stop.

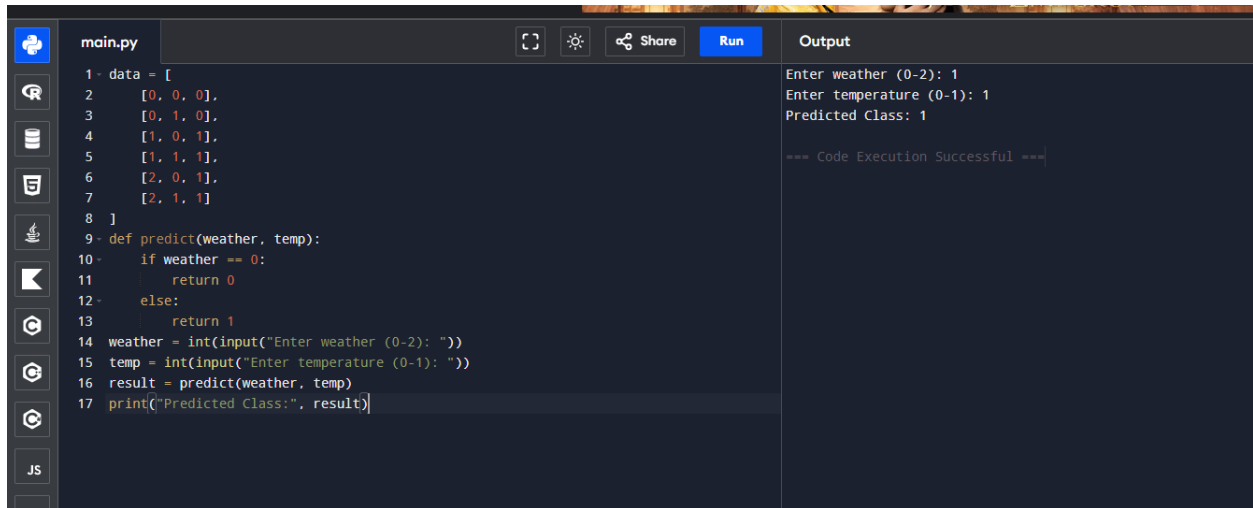
FLOWCHART



PYTHON PROGRAM

```
data = [  
    [0, 0, 0],  
    [0, 1, 0],  
    [1, 0, 1],  
    [1, 1, 1],  
    [2, 0, 1],  
    [2, 1, 1]  
]  
  
def predict(weather, temp):  
    if weather == 0:  
        return 0  
    else:  
        return 1  
  
weather = int(input("Enter weather (0-2): "))  
temp = int(input("Enter temperature (0-1): "))  
result = predict(weather, temp)  
print("Predicted Class:", result)
```

SAMPLE OUTPUT



The screenshot shows a Python IDE with a file named 'main.py'. The code defines a list of training data points, a prediction function, and a main execution block. The output window shows the program's execution, including user input for weather and temperature, and the resulting predicted class.

```
1- data = [  
2-     [0. 0. 0],  
3-     [0. 1. 0],  
4-     [1. 0. 1],  
5-     [1. 1. 1],  
6-     [2. 0. 1],  
7-     [2. 1. 1]  
8- ]  
9- def predict(weather, temp):  
10-     if weather == 0:  
11-         return 0  
12-     else:  
13-         return 1  
14- weather = int(input("Enter weather (0-2): "))  
15- temp = int(input("Enter temperature (0-1): "))  
16- result = predict(weather, temp)  
17- print("Predicted Class:", result)
```

Output:

```
Enter weather (0-2): 1  
Enter temperature (0-1): 1  
Predicted Class: 1  
  
=== Code Execution Successful ===
```

Here:

- 0 → No
- 1 → Yes

RESULT

Thus, the **Decision Tree classification algorithm** was successfully implemented in Python, and the class of new input data was predicted.

CONCLUSION

The experiment demonstrates how a Decision Tree uses **feature-based decisions** to classify data. It is a simple and effective machine learning technique for classification and prediction problems.